MIL-S-19500/591
30 January 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON TYPES 2N7123, 2N7124, 2N7125, 2N7126, JANTX, JANTXV, and JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for N-channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500, with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AR}). Superseding DESC drawing 89007 of 19 December 1989 (see 6.4).

1.2 Physical dimensions. See figure 1 (TO-204AE for types 2N7123 and 2N7124; TO-204AA for types 2N7125 and 2N7126 (formerly TO-3)), see figure 2 for JANC (die) dimensions.

1.3 Maximum ratings (T_A = +25°C, unless otherwise specified).

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C (free air)	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C	I_S	I_{DM} 3/	T_{op} and T_{STG}
	W	W	V dc	A dc	A dc	A dc	A (pk)	°C
2N7123	150	4	±20	38.0	24.0	38.0	152	-55 to +150
2N7124	150	4	±20	30.0	19.0	30.0	120	-55 to +150
2N7125	150	4	±20	14.0	9.0	14.0	56	-55 to +150
2N7126	150	4	±20	12.0	7.75	12.0	48	-55 to +150

Type	E_{AS}	E_{AR}	I_{AR}	V_{ISO} 70,000 ft altitude	Max $r_{DS(on)}$ 4/ V_{GS} = 10 V dc I_D = I_{D2}		$R_{\theta JC}$ max
					T_J = +25°C	T_J = +150°C	
	mJ	mJ	A		ohm	ohm	°C/W
2N7123	150	15	38.0	---	0.055	0.105	0.83
2N7134	500	15	30.0	---	0.085	0.170	0.83
2N7125	700	15	14.0	400	0.300	0.750	0.83
2N7126	750	15	12.0	500	0.400	1.000	0.83

See footnotes on top of next page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5280 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1/ Derate linearly, 1.2 W/°C for T_C > +25°C. $P_T = \frac{T_J \text{ max} - T_C}{R_{\theta JC}}$

2/ $I_D = \sqrt{\frac{T_J \text{ max} - T_C}{(R_{\theta JC}) \times (R_{DS(on) \text{ at } T_J \text{ max}})}}$

3/ $I_{DM} = 4\ I_{D1}$; I_{D1} as calculated in footnote 2.

4/ Pulsed (see 4.5.1).

1.4 Primary electrical characteristics at T_C = 25°C, unless otherwise specified.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V	Max $r_{DS(on)1}$ 1/ V_{GS} = 10 V dc $I_D = I_{D2}$
				V_{DS} = 80% of rated V_{DS}	T_J = +25°C
	V dc	V dc Min	Max	µA dc	ohms
2N7123	100	2.0	4.0	25	0.055
2N7124	200	2.0	4.0	25	0.085
2N7125	400	2.0	4.0	25	0.3
2N7126	500	2.0	4.0	25	0.4

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

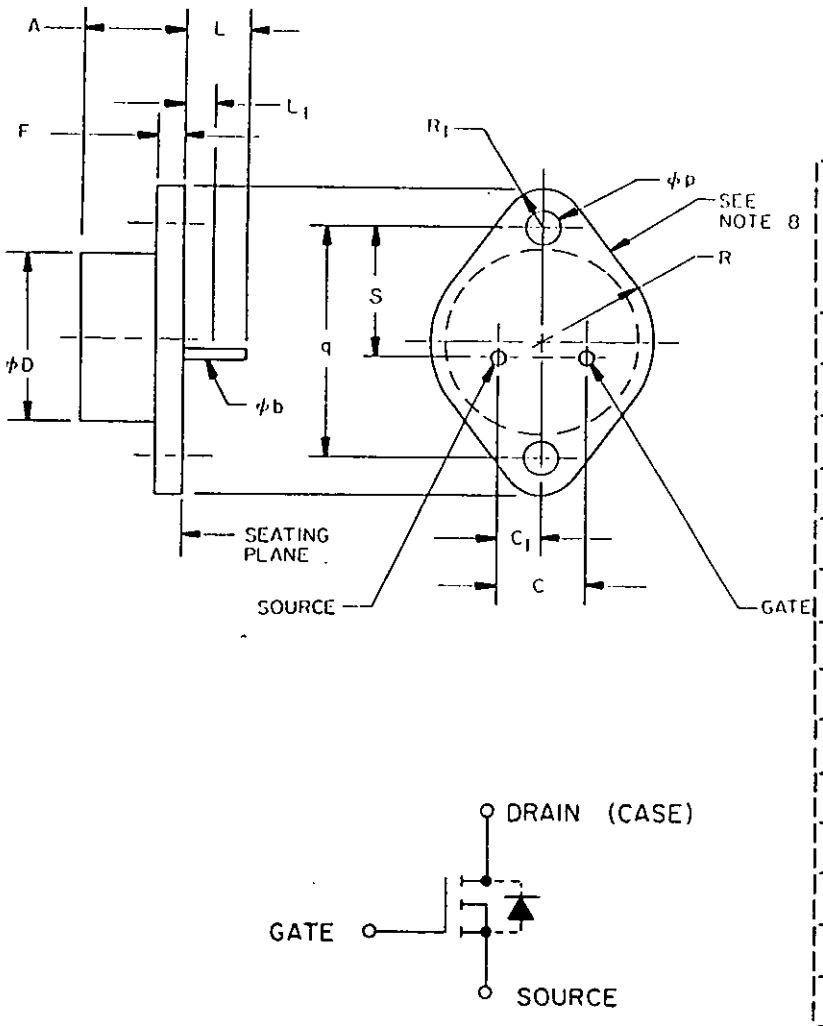
STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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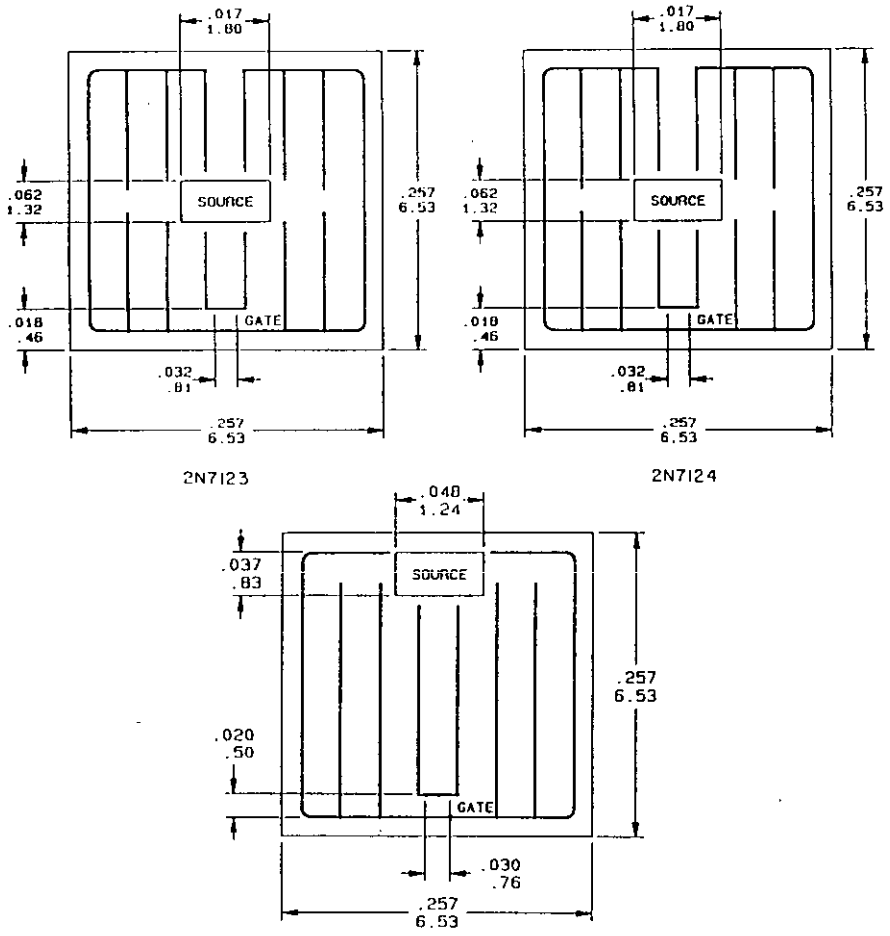


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.360	6.35	9.15	
øb	.057	.063	1.45	1.60	5
øb	.038	.043	0.97	1.10	6
øD	---	.875	---	22.23	
C	.420	.440	10.67	11.18	7
C_1	.205	.225	5.21	5.72	7
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1	---	.050	---	1.27	
øp	.151	.165	3.84	4.09	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. These dimensions pertain to the TO-204AE package types.
6. These dimensions pertain to the TO-204AA package types.
7. Mounting holes shall be deburred on the seating plane side.
8. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions.

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Inches	mm	Inches	mm	Inches	mm
.016	0.41	.031	0.79	.116	2.95
.018	0.46	.033	0.84	.148	3.76
.019	0.48	.034	0.86	.180	4.57
.025	0.64	.041	1.04	.181	4.60
.026	0.66	.106	2.69		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The physical characteristics of the die thickness are .0187 (0.474 mm). The back metals are chromium, nickel and silver. The top metal is aluminum and the back contact is the drain.
5. Die dimensions and characteristics for International Rectifier (CAGE code 59993)

FIGURE 2. JANC die dimensions.

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 and 2 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-S-19500, and herein. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.5).

3.3.2 Internal construction. Multiple chip construction is not be permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed.

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500, and table IV herein.

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4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

4.3 1 Screening (JANC). Screening shall be in accordance with MIL-S-19500 (appendix H). As a minimum die shall be 100 percent probed per group 2 except test current shall not exceed 20 A.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/ 2/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/ 2/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein	Subgroup 2 of table I herein 3/
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$ $V_{GS(th)1}$ subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$ $V_{GS(th)1}$ subgroup 2 of table I herein
12	Method 1042, test condition A, t = 240 hours	Method 1042, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater $\Delta r_{DS(on)1}$ = ±20% of initial value $\Delta V_{GS(th)1}$ = ±20% of initial value	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100% of initial value, whichever is greater ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater $\Delta r_{DS(on)1}$ = ±20% of initial value $\Delta V_{GS(th)1}$ = ±20% of initial value

1/ Shall be performed anytime before screen 10.
2/ This is a stress test designed to ensure a rugged product.
3/ These tests shall be performed after methods 3470, 3161, and gate stress and shall precede screen 10.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and table IIa and IIb herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC}$ max = 0.83°C/W.

a.	I_M measuring current	10 mA
b.	I_H drain heating current	3 A
c.	t_H heating time	Steady-state (see method 3161 of MIL-STD-750, for definition)
d.	V_H drain source heating voltage	25 V
e.	t_{MD} measurement time delay	30 to 60 µs
f.	t_{SW} sample window time	10 µs (max)

4.5.3 Thermal response (ΔV_{SD} measurement). The delta V_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 3). The read and record delta V_{SD} measurements and conditions for each device in the qualification lot shall be submitted (read and record) in the qualification report. The chosen delta V_{SD} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply.

a.	I_M measuring current	10 mA
b.	I_H drain heating current	3 A (min)
c.	t_H heating time	100 ms
d.	V_H drain source heating voltage	25 V
e.	t_{MD} measurement time delay	30 to 60 µs
f.	t_{SW} sample window time	10 µs (max)

4.5.4 Single pulse avalanche energy E_{AS}.

a.	Peak current	I_{D1}
b.	Peak gate voltage, V_{GS}	10 V
c.	Gate to source resistor	$25 \leq R_{GS} \leq 200\Omega$
d.	Initial case temperature	+25°C +10, -5°C
e.	Inductance	$\left[2\ E_{AS}/(I_{D1})^2\right]\left[(V_{BR} - V_{DD})/V_{BR}\right]$ mH minimum
f.	Number of pulses to be applied	1 pulse minimum
g.	Supply voltage	50 V

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum.
b. t = 250 µs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1 mA dc Condition C	$V_{(BR)DSS}$			V dc
2N7123				100	---	
2N7124				200	---	
2N7125				400	---	
2N7126				500	---	
Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc and -20 V dc Bias condition C, V_{DS} = 0	I_{GSS1}	---	±100	nA dc
Drain current	3413	V_{GS} = 0 V dc, V_{DS} = 80% of rated V_{DS}, Bias condition C	I_{DSS1}	---	25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Pulsed (see 4.5.1) Condition A, I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			Ω
2N7123				---	0.055	
2N7124				---	0.085	
2N7125				---	0.3	
2N7126				---	0.4	
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Pulsed (see 4.5.1) Condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			Ω
2N7123				---	0.065	
2N7124				---	0.09	
2N7125				---	0.40	
2N7126				---	0.50	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V dc, I_D = I_{D1}	V_{SD}			V
2N7123					1.9	
2N7124					1.9	
2N7125					1.7	
2N7126					1.7	
Subgroup 3						
High temperature operation		T_C = T_J = +125				
Gate current	3411	Bias condition C V_{GS} = +20 V dc and -20 V dc V_{DS} = 0 V dc	I_{GSS2}	---	±200	nA dc
Drain current	3413	Bias condition C V_{GS} = 0 V dc V_{DS} = 100% of rated V_{DS} V_{DS} = 80% of rated V_{DS}	I_{DSS2} I_{DSS3}	--- ---	1.0 0.25	mA dc mA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Pulsed (see 4.5.1) I_D = rated I_{D2}	$r_{DS(on)3}$			Ω
2N7123				---	0.094	
2N7124				---	0.153	
2N7125				---	0.66	
2N7126				---	0.88	
Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)2}$	1.0	---	V dc
Low temperature operation		T_C = T_J = -55°C				
Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)3}$	---	5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1} V_{GS} = 10 V dc Gate drive Impedance = 2.35Ω V_{DD} = 0.5 $V_{BR(DSS)}$				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Turn-on delay time			$t_{d(on)}$	---	35	ns
2N7123						
2N7124						
2N7125						
2N7126						
Rise time			t_r		190	ns
2N7123				---		
2N7124				---		
2N7125				---		
2N7126				---		
Turn-off delay time			$t_{d(off)}$		170	ns
2N7123				---		
2N7124				---		
2N7125				---		
2N7126				---		
Fall time			t_f		130	ns
2N7123				---		
2N7124				---		
2N7125				---		
2N7126				---		
Subgroup 5						
Safe operating area test	3474	See figure 4, V_{DS} = 80% of rated $V_{BR(DSS)}$ t_p = 10 ms, V_{DS} = 200 V max				
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			nC
2N7123				50	125	
2N7124				55	115	
2N7125				52	110	
2N7126				55	120	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Gate to source charge			Q_{gs}			nC
2N7123				8	22	
2N7124				8	22	
2N7125				5.0	18	
2N7126				5.0	19	
Gate to drain charge			Q_{gd}			nC
2N7123				25	65	
2N7124				30	60	
2N7125				25	65	
2N7126				27	70	
Reverse recovery time	3473	di/dt = 100 A/µs $V_{DD} \leq$ 30 V dc $I_D = I_{D1}$	t_{rr}			ns
2N7123				---	500	
2N7124				---	950	
2N7125				---	1200	
2N7126				---	1600	

1/ For sampling plan, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS device.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2 2/		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 3		
Temperature cycling	1051	Test condition G
Hermetic seal	1071	
Fine leak Gross leak		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7
Decap internal visual (design verification)	2075	See 3.3.2
SEM	2077	
Bond strength (wire or clip bonded)	2037	Test condition A: All internal wires for each device shall be pulled separately.
Die shear	2017	
Subgroup 4		
Intermittent operation life (LTPD)	1042	Condition D: The heating cycle shall be 1 minute minimum, 2,000 cycles. No heat sink nor forced air cooling on the device shall be permitted, during the on cycle.
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, and 8.

See footnotes at end of table.

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TABLE IIa. Group B inspection for JANS device - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 5 2/		
Accelerated steady-state reverse bias	1042	Condition A, V_{DS} = rated T_A = +175°C, t = 120 hours Read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1 mA Read and record I_{DSS} (pre and post) in accordance with table V.
Electrical measurements		See table V herein, steps 1, 2, 3, 4, 5, 6, and 7. No more than 15% of the sample shall be permitted to have a ΔV_{DS} shift of more than 10% and ΔI_{DSS} greater than 50 µA.
Accelerated steady-state gate stress		Condition B, V_{GS} = rated, T_A = +175 °C, t = 24 hours.
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7.
Bond strength (Al-Au die interconnects only)	2037	Test condition A
Subgroup 6		
Thermal impedance	3161	See 4.5.2

1/ For sampling plan, see MIL-S-19500.
2/ A separate sample may be pulled for each test.

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TABLE IIb. Group B inspection for JANTX and JANTXV.

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1 2/		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Temperature cycling	1051	Condition G
Hermetic seal	1071	
Fine leak Gross leak		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7
Subgroup 3		
Intermittent operation	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum.
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, and 8
Subgroup 4		
Decap internal visual (design verification)	2075	See 3.3.2
Bond strength (wire or clip bonded)		Test condition A. All internal bond wires for each device shall be pulled separately.
Subgroup 5		
Not applicable		
Subgroup 6		
High-temperature life (nonoperating)	1032	T_A = +150°C
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7

1/ For sampling plan, see MIL-S-19500.
2/ A separate sample may be pulled for each test.

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TABLE III. Group C inspection (all quality levels).

Inspection 1/	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Thermal shock (glass strain)	1056	Condition A
Terminal strength	2036	Test condition A: Weight = 10 lbs, t = 15 seconds
Hermetic seal	1071	
Fine leak Gross leak		
Moisture resistance	1021	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	
Constant acceleration	2006	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7
Subgroup 4		
Salt atmosphere	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Intermittent operation	1042	Test condition D: 6,000 cycles. The heating cycle shall be 1 minute minimum.
Hermetic seal	1071	
Fine leak Gross leak		
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, and 8

1/ For sampling plan, see MIL-S-19500.

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TABLE IV. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			45 devices c = 1
Temperature cycling	1051	Condition G, 500 cycles	
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, 7, and 8	
Subgroup 2 1/			45 devices c = 1
Steady-state reverse bias	1042	Condition A: 1,000 hours	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7	
Steady-state gate bias	1042	Condition B: 1,000 hours	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			3 devices, c = 0
Destructive physical analysis (DPA)	2102		c = 0
Subgroup 4			5 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.83°C/W max. See 4.5.2	
Subgroup 5			5 devices c = 0
Barometric pressure (reduced) 400 and 500 V only	1001	Test condition C, $V_{ISO} = V_{DS}$ $I_{(ISO)}$ = .25 mA (max)	
Subgroup 6			5 devices c = 0
Repetitive avalanche energy	3469	Peak current $I_{AR} = I_D$ Peak gate voltage V_{GS} = 10 V Gate to source resistor = 25 $\leq R_{GS} \leq$ 200Ω Temperature = T_J = 150°C +0 -10°C Inductance = $[2 E_{AR}/(I_{D1})^2]$ $[(V_{BR} - V_{DD})/V_{BR}]$ mH minimum Number of pulses to be applied = 3.6×10^8 Supply voltage (V_{DD}) = 50 V Time in avalanche = 2 µs minimum 20 µs maximum Frequency = 500 Hz minimum	

1/ A separate sample for each test may be pulled.

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TABLE V. Groups A, B, C, and E electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source	3407	V_{GS} = 0, I_D = 1 mA dc Bias condition C	$V_{(BR)DSS}$			V dc
	2N7123				100	---	
	2N7124				200	---	
	2N7125				400	---	
	2N7126				500	---	
2.	Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
3.	Gate current	3411	V_{GS} = ±20 V dc Bias condition C	I_{GSS1}	---	±100	nA dc
4.	Drain current	3413	V_{GS} = 0, V_{DS} = 80% of rated V_{DS}, bias condition C	I_{DSS1}	---	25	µA dc
5.	Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Condition A Pulsed (see 4.5.1) $I_D = I_{D2}$	$r_{DS(on)1}$			Ω
	2N7123				---	0.055	
	2N7124				---	0.085	
	2N7125				---	0.3	
	2N7126				---	0.4	
6.	Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Condition A Pulsed (see 4.5.1) $I_D = I_{D1}$	$r_{DS(on)2}$			Ω
	2N7123				---	.065	
	2N7124				---	.090	
	2N7125				---	.400	
	2N7126				---	.500	
7.	Forward voltage (source-drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V dc $I_D = I_{D1}$	V_{SD}			V
	2N7123					1.9	
	2N7124					1.9	
	2N7125					1.7	
	2N7126					1.7	
8.	Thermal response	3161	See 4.5.3	ΔV_{SD}			

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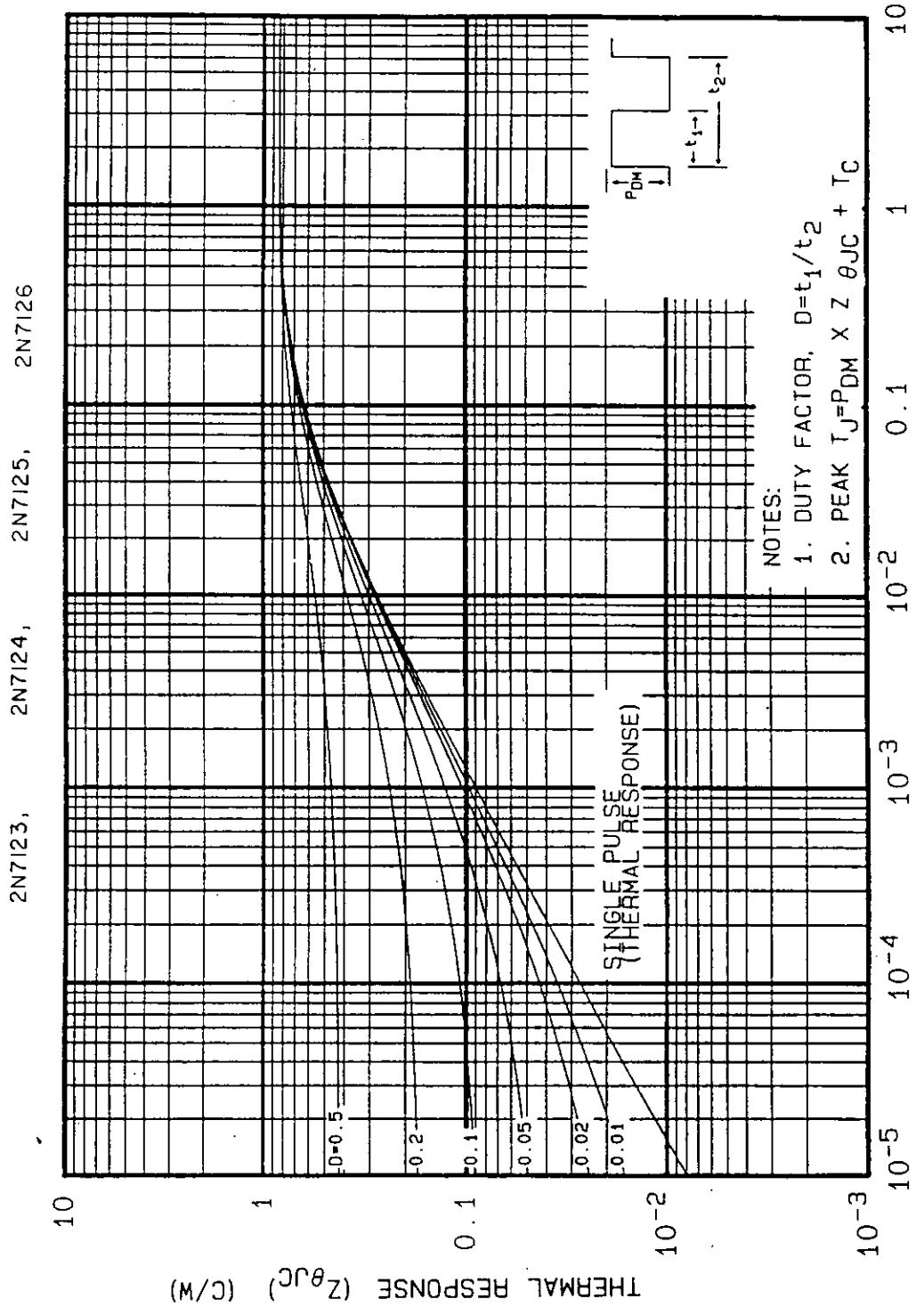
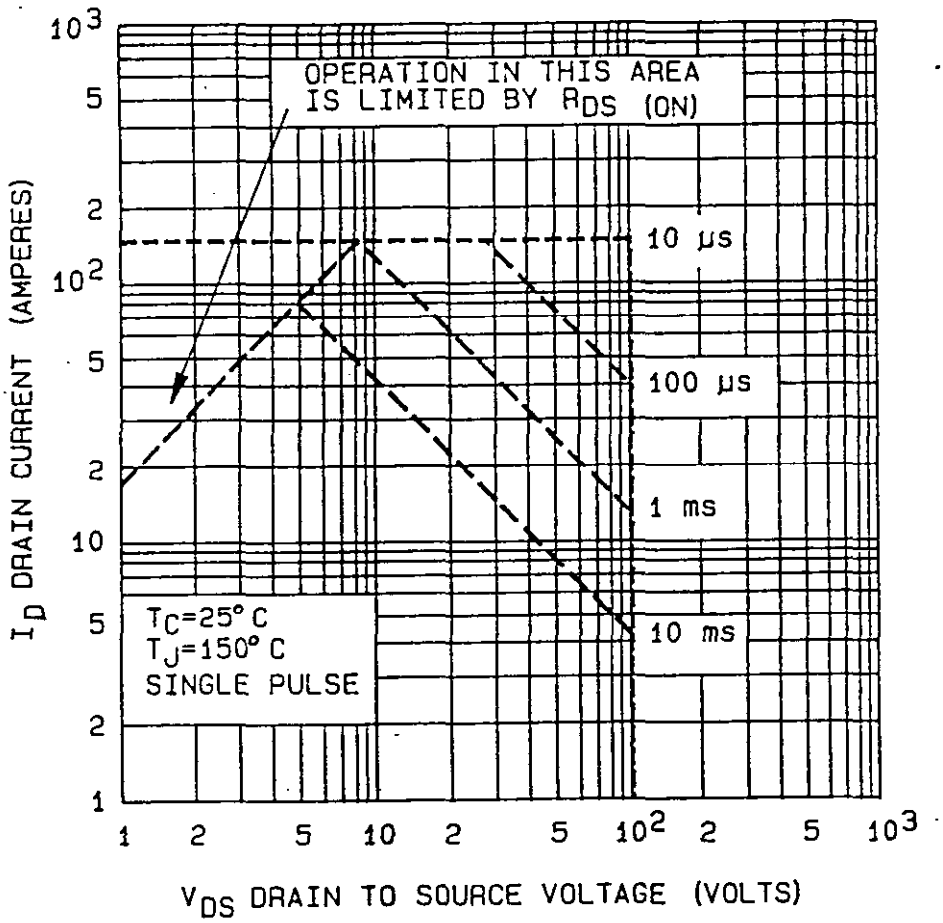


FIGURE 3. Thermal response curves.

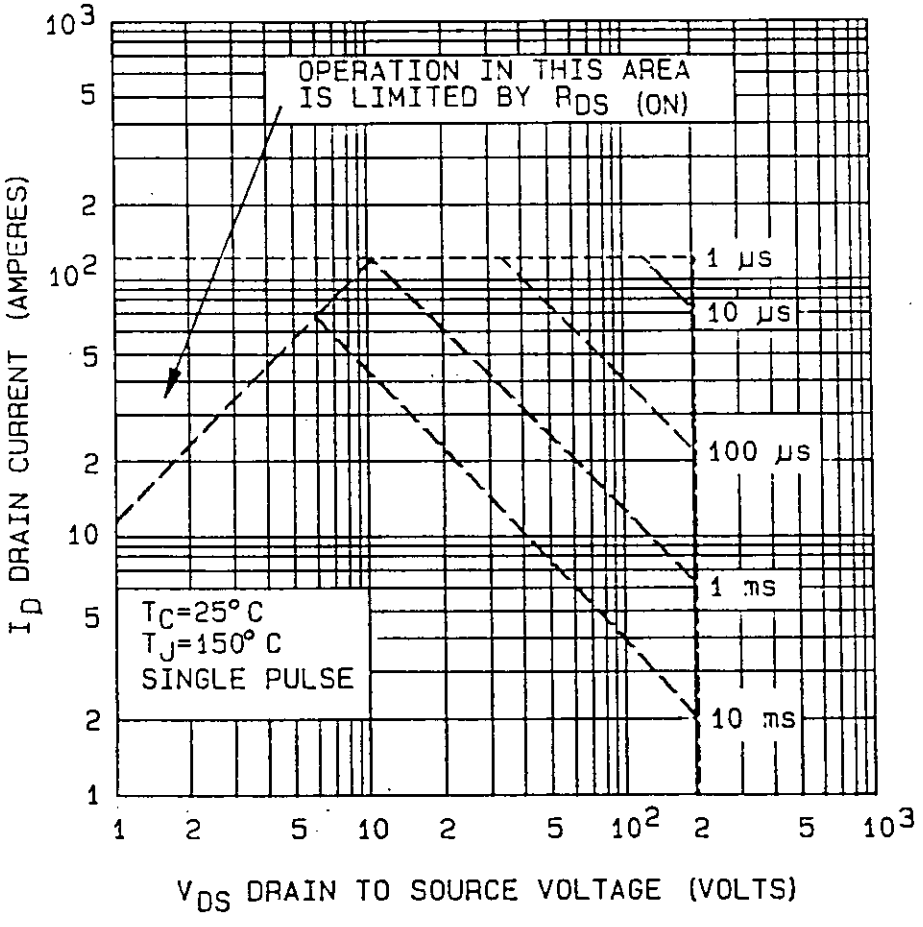
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2N7123

FIGURE 4. *Maximum safe operating area.*

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2N7124

FIGURE 4. Maximum safe operating area - Continued.

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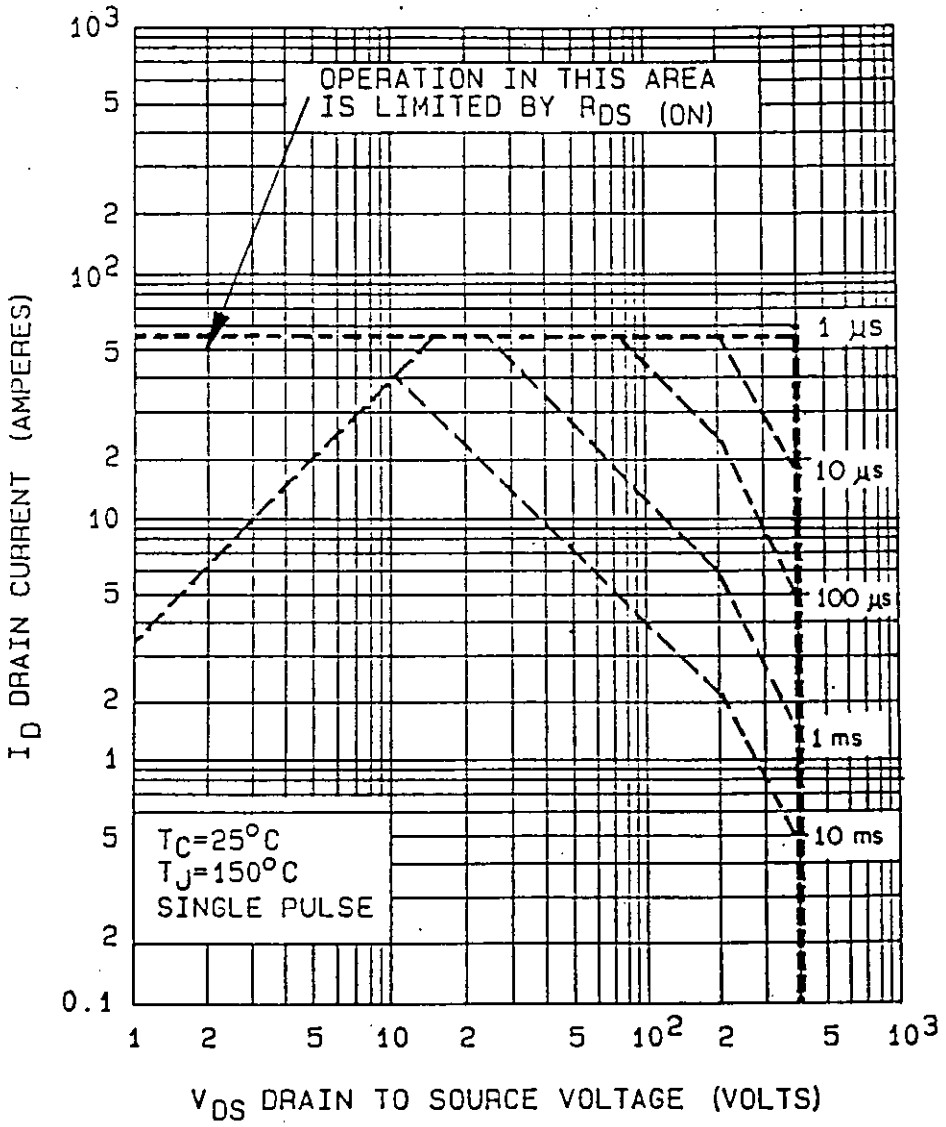


FIGURE 4. Maximum safe operating area - Continued.

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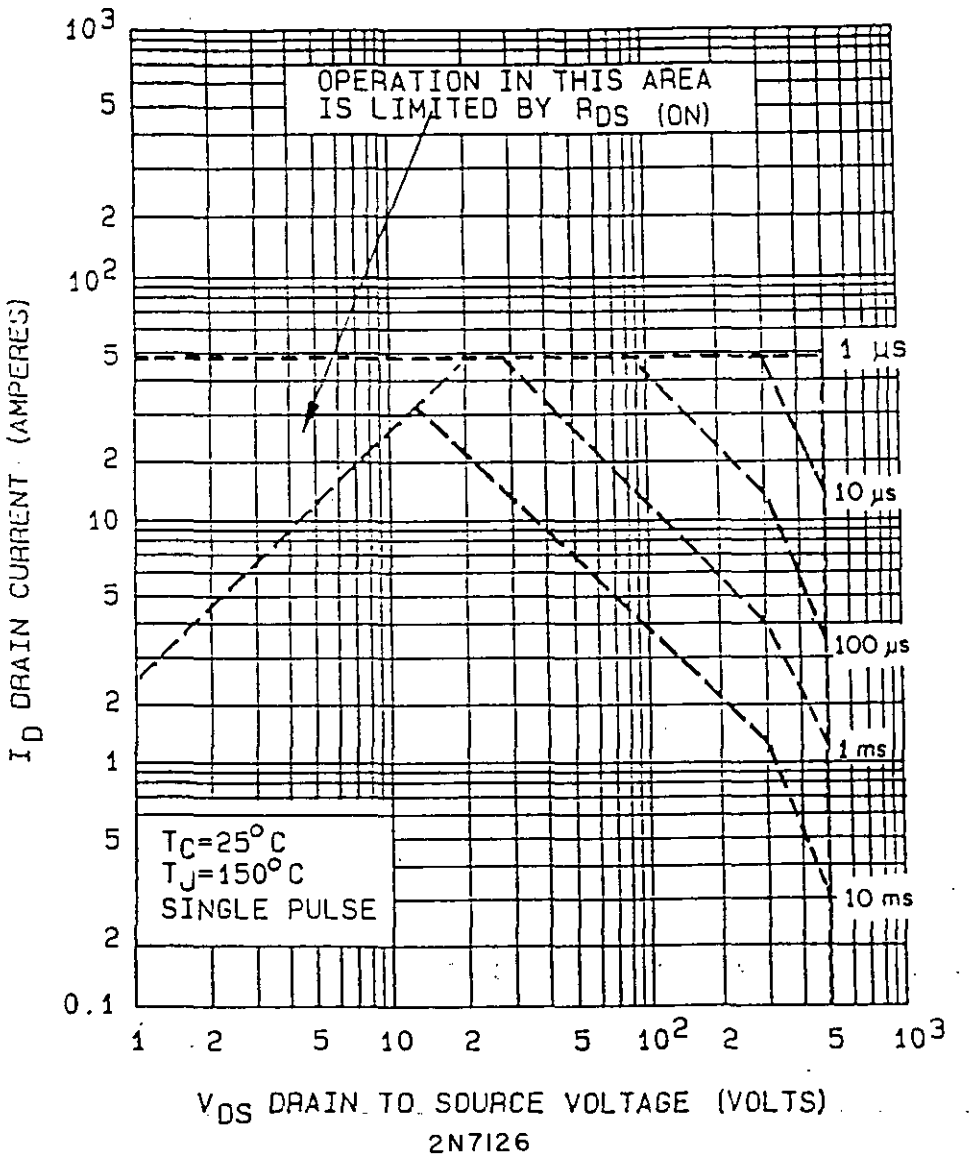


FIGURE 4. Maximum safe operating area - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead material and finish may be specified (see 3.3.1).

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part number. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military Part or Identifying Number (PIN).

PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
2N7123	59993, 18722	IRF150
2N7124	59993, 18722	IRF250
2N7125	59993, 18722	IRF350
2N7126	59993, 18722	IRF450

6.4 Supersession data. This specification supersedes DESC drawing 89007 of 19 December 1989 (see 1.1), upon the first qualification completion.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 70, 80
DLA - ES

User activity:
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1262)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-19500/591	91/01/30

3. DOCUMENT TITLE
Semiconductor Device, Repetitive Avalanche, Field Effect, Transisitor, N-Channel

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
	(1) Commercial	(2) AUTOVON
ALAN BARONE	513-296-6048	986-6048
c. ADDRESS (Include Zip Code) DESC-ECT Dayton, OH 45444-5280	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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